

A model predictive control strategy of PV-Battery microgrid under variable power generations and load conditions

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HIGHLIGHTS

- A state of charge (SOC)-oriented charging scheme for battery.
- A model predictive control for interlinking converter.
- Energy management system for stable operation of the PV-Battery microgrid.
- Exploit the microgrid reactive capability for grid support.

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ABSTRACT

Renewable energy sources have been increasingly deployed as distributed generators in remote areas. Meanwhile, fluctuating power generation from renewable energy sources, together with variable power demand, poses challenges in stable and reliable power supply. In this paper, a microgrid with solar photovoltaic (PV) and battery energy storage (BES) is studied. A state of charge (SOC)-oriented charging scheme is developed to control the BES to smooth the PV output. Most importantly, a sophisticated control algorithm, consisting of a model predictive voltage control (MPVC) and a model predictive power control (MPPC), is proposed for the interlinking converter. It enables stable voltage in islanded mode. Also, in grid-connected mode, flexible reactive power can be injected into the main grid for grid support according to the voltage variation level. Finally, by considering the intermittent nature of the PV and the load profile, an energy management system (EMS) is designed to ensure power balance within the system. Case studies are provided to demonstrate the effectiveness of the proposed control strategy.

1. Introduction

With growing concern on the environmental issues and the rapid depletion of traditional fossil fuel, electrical grids tend to be more distributed, intelligent, and flexible with high penetration of renewable energy sources. Electrical and energy engineering now have to face a new scenario in which small distributed power generators and dispersed energy-storage devices have to be integrated together into microgrids [1,2]. In order to successfully integrate renewable distributed energy resources, many technical challenges must be overcome to ensure that the present levels of reliability are not significantly affected, and the potential benefits of distributed generation are fully harnessed [3].

In the control technique development for various microgrids, the

energy sources are usually replaced by ideal DC sources by many researchers [4–7]. This is because the simulation of real renewable energy sources by using ideal DC sources can simplify the system and hence, facilitate the control strategies development. However, the intermittent nature of the renewable energy sources cannot be ignored in practical microgrids. Under such practical considerations, original control approaches may not work well when demanded power cannot be met by energy sources due to intermittency.

To avoid such occurrence, suitable microgrid configurations and sophisticated control are necessary by considering both varying power generation and consumption. Battery energy storage (BES) plays a significant role in smoothing the renewable energy output [8,9]. It has been promoted as the “game-changer” worldwide in the development of future grids with capability of keeping the power balance and

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maintaining voltages. In [10], a BES system is employed to smooth the PV output and increase the level of the microgrid scalability based on a decentralized control approach. In [11] and [12], the BES is used to stabilize the wind-PV microgrid and supply stable power for the loads. In such microgrid systems, however, the grid-connected operation is not developed. This, to a large extent, limits their applications.

Microgrids with multiple energy sources and ac-dc subgrids that can operate in either grid-tied or autonomous modes have been studied in [13–16]. In such systems, the interlinking converter plays an important role in ensuring power balance. In autonomous mode, the ac/dc interlinking converter is controlled to provide a stable ac voltage and frequency for the ac grid while the dc-bus voltage is maintained by the bidirectional dc-dc converter [13]. In grid-tied mode, the ac/dc interlinking converter is used to maintain a stable dc-bus voltage and to exchange power between the microgrid and the utility [14]. In the literatures mentioned above, conventional multiple feedback loops with proportional-integral-differential (PID) controllers are commonly used to control the interlinking converters [16]. Conventional cascade linear controllers suffer from PID parameters tuning work and low dynamic response, due to their cascaded structures. Besides, complicated co-ordination transformation and PWM modulation are needed. Also, different control structures are required for grid-tied and autonomous operations.

Even grid-connected operation has also been achieved, another concern is raised from the operation of BES. In most of the research work mentioned above, the BES is kept discharging or being charged with constant rate in grid-connected mode. This is obviously not an appropriate approach as overcharge or overdischarge may occur without considering the state-of-charge (SOC). By using SOC feedback and real-time power allocation, a control strategy was developed for smoothing the fluctuating output of the renewable energy [17]. The SOC can be regulated within a specified range while the BES units are able to share the load consistently. The details about BES operation in grid-connected mode is however not investigated. It is noted that modeling uncertainties and current/voltage sensor bias may cause estimated SOC drift, which in turn, can deteriorate the system performance. To obtain accurate SOC estimation, a variety of closed-loop techniques have been developed. For instance, adaptive filters such as Luenberger observer [18] and Kalman Filter [19] are employed due to their low computational complexity and reasonable accuracy. More recently, a recursive total least squares (RTLS) technique based on Rayleigh quotient minimization is developed to eliminate unexpected sensing noises during SOC estimation [20].

Another aspect that is seldom mentioned in the existing microgrid control is the reactive capacity for grid support. As reactive power can directly regulate the grid voltage, reactive power support is usually provided by capacitor banks, static Var compensators (SVCs) or static synchronous compensators (STATCOMs) in a centralized manner [21,22]. This is the most common method adopted by present grid operators. However, recent research has revealed that localized control can achieve nearly 80% saving in losses when compared to a centralized control [23]. In the future grid, more and more microgrids are integrated into the distribution network. Therefore, it may be efficient to provide grid support near the consumers by flexibly controlling the reactive powers of the microgrids [24].

To address the issues mentioned above, a model predictive control (MPC) strategy is developed in this paper. Actually some control algorithms of power flow using MPC have been proposed for microgrids to achieve a variety of aims such as minimizing system operating costs [25], economic load dispatch [26], energy storage [27], optimized power flow management [28], to name but a few. These algorithms are designed and implemented only at the system level while the structures of the microgrids and the power converters are not considered. For the first time in ac/dc microgrids consisting of renewable energy sources and energy storage, here the MPC is applied to control the interlinking converter to achieve multiple objectives including stable voltage

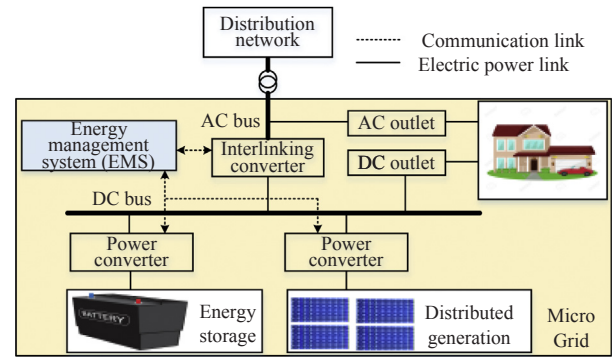


Fig. 1. Proposed hybrid ac/dc microgrid structure.

supply, flexible power regulation and grid support.

2. Microgrid configuration and modeling

2.1. Microgrid configuration

Fig. 1 shows the microgrid architecture. A solar PV array is connected to the dc bus through a dc/dc boost converter. A battery as energy storage is connected to the same dc bus via a bidirectional dc/dc converter. The ac and dc buses are interconnected through a bidirectional ac/dc interlinking converter. Variable dc and ac loads are connected to the dc and ac buses, respectively. This hybrid ac/dc microgrid can be further connected to the distribution network through a static transfer switch (STS). The control center collects the real-time data and delivers the control demands through the communication link. The power converters coordinately control the power flow based on an energy management system (EMS) through the electric power link. Even although it is out of the scope of this work, it is worth mentioning that these small microgrids can be interconnected to each other through the distribution network operator (DNO) under the future smart grid framework. The global central controller (GCC) is to coordinately control the microgrids for flexible power sharing [29].

In this microgrid, the mathematical models of the PV, battery and the ac/dc interlinking converter are essential for the control technique development. These models are now well known but for completeness they are quoted below.

2.2. Modeling of PV system

The PV system is shown in Fig. 2. The PV array is connected to the common dc bus through a boost dc-dc converter. The PV array consists of SunPower SPR-305-WHT solar modules. Solar modules are connected in series to form strings that are further connected in parallel as an array. The solar module can be equivalent to a current source in parallel with a diode [30]. Fig. 3 depicts the output characteristics of the PV. Under different solar irradiance, the maximum power points of the power-voltage curves are associated with different output voltages. Also, under certain solar irradiance, the output of the PV panel is

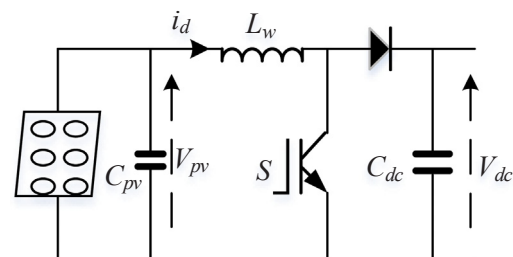


Fig. 2. PV system configuration.

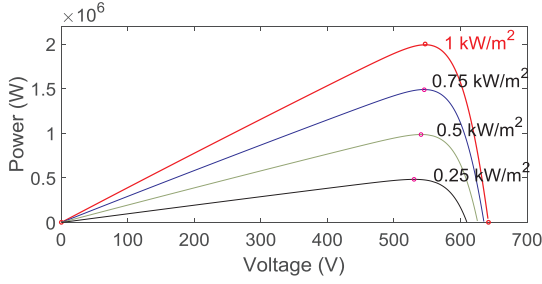


Fig. 3. Output characteristics of the PV array.

varying with different terminal voltages.

As explained previously, to maximize the utilization of solar energy from the PV farm, solar PV system should be operated in maximum power point tracking (MPPT). This can be achieved by adjusting the PV output voltage by controlling the boost converter. In autonomous mode, the boost converter can be operated in on-MPPT or off-MPPT according to the power balance of the system and the SOC of the battery. The MPPT techniques of solar PVs have been widely developed. In this paper, an incremental conductance method [31] is used. The PV system performance is shown in Fig. 4. The solar irradiance drops down to 250 W/m² gradually and then ramps up to 1000 W/m² again. The temperature is kept at 25 °C. The MPPT algorithm starts to operate at 0.4 s. It can be seen that the PV system is continuously adjusting the optimal voltage under solar irradiance variation. As a result, the output power of the PV closely follows the solar irradiation with maximum power output.

2.3. Modeling of battery energy storage (BES)

Due to the intermittent nature of renewable energy resources, energy storage systems are critical to smooth the power gap within the microgrid. In this paper, the battery is connected to the dc bus through a bidirectional buck-boost converter for charging or discharging, as shown in Fig. 5. Two important parameters to represent a battery are terminal voltage and the SOC as follows [32].

$$V_{Bat} = V_o - R_b i_b - K \frac{Q}{Q - \int i_b dt} + A e^{(-B \int i_b dt)} \quad (1)$$

$$SoC = 100 \left(1 - \frac{\int i_b dt}{Q} \right) \quad (2)$$

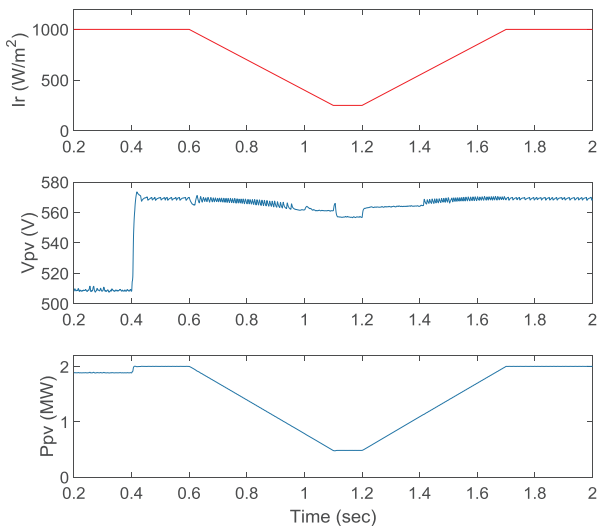


Fig. 4. PV output power versus solar irradiation.

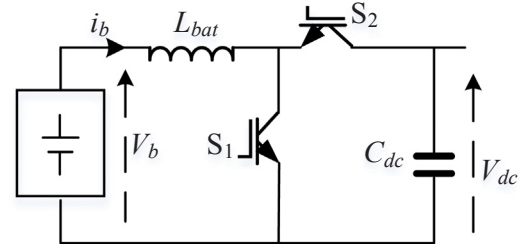


Fig. 5. Energy storage system.

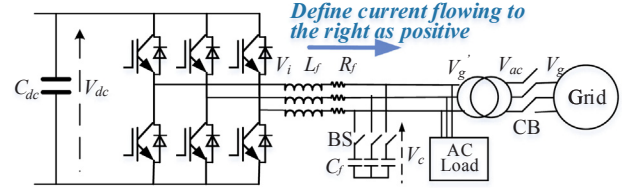


Fig. 6. AC side of the microgrid.

where R_b is internal resistance of the battery, V_o is the open circuit voltage of the battery, i_b is battery discharging current, K is polarization voltage, Q is battery capacity, A is exponential voltage, and B is exponential capacity.

2.4. Modeling of the interlinking converter

Fig. 6 describes the ac side of the microgrid. The ac/dc interlinking converter is the main power converter of the system. It not only acts as an interface for smooth power transfer between dc subgrid and ac subgrid, but also can be controlled for flexible active and reactive powers regulation between the microgrid and the utility. In grid-tied mode, the circuit breaker (CB) is turned ON and the bypass switch (BS) is switched OFF. The mathematical model of the AC side can be expressed as

$$\mathbf{V}_i = \mathbf{V}'_g + \mathbf{I}_f R_f + L_f \frac{d\mathbf{I}_f}{dt} \quad (3)$$

where $\mathbf{V}_i$ and $\mathbf{V}'_g$ are the converter voltage vector and the grid voltage vector referred to the microgrid side, respectively; $\mathbf{I}_f$ the inverter output current vector; L_f the filter inductance; R_f the equivalent resistance. The output active and reactive powers of the ac/dc interlinking converter are

$$P = \frac{3}{2} \text{Re}\{\mathbf{V}'_g \mathbf{I}_f^*\} = \frac{3}{2} (V'_{g\alpha} I_{f\alpha} + V'_{g\beta} I_{f\beta}) \quad (4)$$

$$Q = \frac{3}{2} \text{Im}\{\mathbf{V}'_g \mathbf{I}_f^*\} = \frac{3}{2} (V'_{g\beta} I_{f\alpha} - V'_{g\alpha} I_{f\beta}) \quad (5)$$

In autonomous operation, the CB is turned OFF and the BS is switched ON. In this case, the dynamic behavior of the capacitor of the LC filter can be expressed as

$$C_f \frac{d\mathbf{V}_c}{dt} = \mathbf{I}_f - \mathbf{I}_L \quad (6)$$

where $\mathbf{V}_c$ is the capacitor voltage vector (i.e. the load voltage vector), C_f the filter capacitance, $\mathbf{I}_L$ the load current vector. The mathematical model of the ac/dc interlinking converter can be described as

$$\mathbf{V}_i = \mathbf{I}_f R_f + L_f \frac{d\mathbf{I}_f}{dt} + \mathbf{V}_c \quad (7)$$

3. SOC-oriented control of BES

The bidirectional dc-dc converter is used to inject power to, or

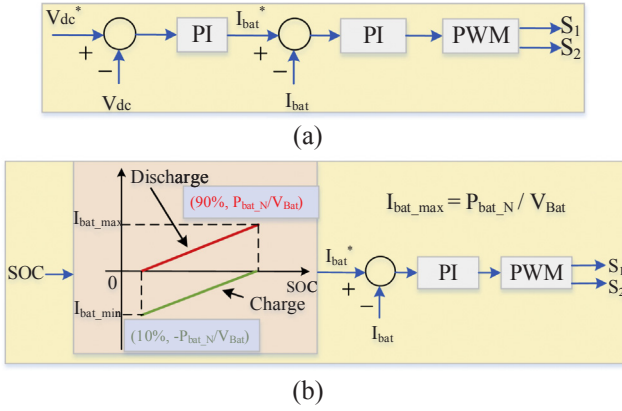


Fig. 7. Battery control scheme. (a) Charging/discharging scheme in autonomous mode, (b) Proposed SOC-oriented scheme in grid-tied mode.

absorb power from the dc bus. In autonomous mode, the bi-directional converter plays an important role to maintain the dc-bus voltage. Usually, a two-loop control is used [14], as depicted in Fig. 7(a). The error between the measured dc-bus voltage and the reference is delivered to a PI controller. The output of this PI controller is set as the charging/discharging current reference for the inner current loop. For instance, when the dc-bus voltage is greater than the reference, the outer voltage controller generates a negative current reference. The inner current loop then adjusts the duty cycle to force the current flow from the dc bus to the battery, which results in charging the battery. Subsequently, the dc bus voltage decreases to the reference as the excess energy is absorbed by the battery. On the contrary, if the dc bus voltage is lower than the nominal value, a positive current reference is produced. Extra energy is provided by the battery to maintain the dc-bus voltage.

In grid-tied mode, the battery can be charged or discharged. Fig. 7(b) illustrates the proposed SOC-oriented charging and discharging scheme. Different to the conventional methods with constant charging/discharging rate, here the charging/discharging rate (i.e., the current reference) varies with the SOC. The red line is the discharging curve and the green line represents the charging characteristic. In this way, the battery can be charged or discharged in a smart manner. Take the charging mode for example, the lower the SOC is, the faster the charging rate becomes. When the SOC reaches the maximum value, I_{bat} become zero to avoid overcharge. After the current reference is determined, it is compared with the measured current. The error between the reference and the measurement is delivered to a PI controller. Finally, the PWM signals are generated according to the PI output to regulate the current flow.

It is noted that, in both operation modes, the SOC limits and the nominal charging/discharging rates should be considered as the energy constraints of the battery

$$10 \leq SOC \leq 90 \quad (8)$$

$$-\frac{P_{Bat_N}}{V_{Bat}} \leq I_{bat} \leq \frac{P_{Bat_N}}{V_{Bat}} \quad (9)$$

where P_{Bat_N} is the rated power of the battery, V_{Bat} is the normal voltage.

4. Proposed MPC strategy of interlinking converter

To control the ac/dc interlink converter, a flexible model predictive control strategy is developed. To be specifically, a model predictive voltage control (MPVC) scheme is developed for autonomous operation, while a model predictive power control (MPPC) scheme is developed for grid-tied mode. Fig. 8 illustrates the proposed control strategy.

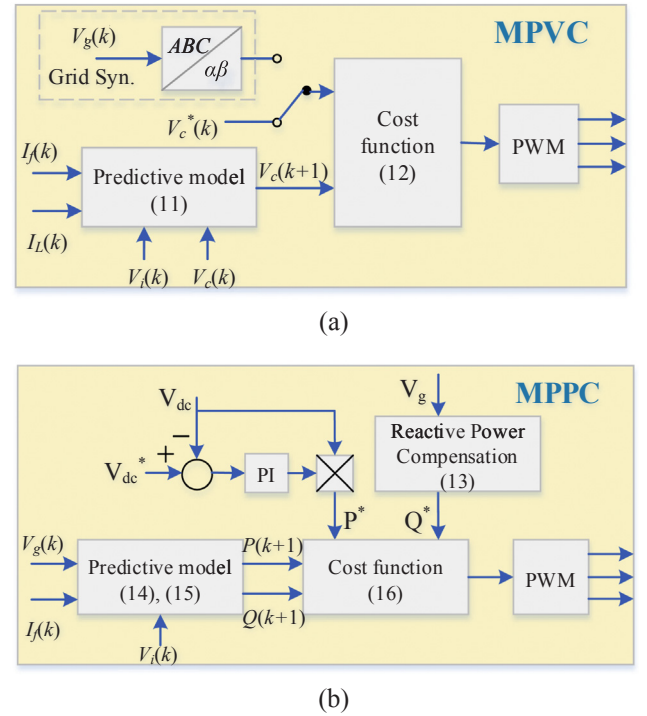


Fig. 8. Proposed control strategy of ac/dc interlinking converter for microgrids. (a) MPVC for autonomous mode, (b) MPPC for grid-tied mode.

4.1. Model predictive voltage control in autonomous mode

Fig. 8(a) shows the block diagram of autonomous mode control. In autonomous operation, the dc-bus voltage can be maintained by controlling the battery bidirectional buck-boost converter, while the dc/ac interlinking converter aims to provide a stable ac voltage for the ac load. With this goal, the capacitor voltage should be the control objective of the MPC controller. Combining (6) and (7), the system model can be rewritten as a state-space system

$$\frac{dx}{dt} = Ax + By \quad (10)$$

where

$$x = \begin{bmatrix} V_c \\ I_f \end{bmatrix}, y = \begin{bmatrix} V_g \\ I_L \end{bmatrix}, A = \begin{bmatrix} 0 & 1/C_f \\ -1/L_f & -R_f/L_f \end{bmatrix}, B = \begin{bmatrix} 0 & -1/C_f \\ 1/L_f & 0 \end{bmatrix}$$

The space-state system in (10) can be expressed in discrete-time as follows:

$$x(k+1) = e^{T_s A} x(k) + A^{-1}(e^{T_s A} - I_{2 \times 2}) B y(k) \quad (11)$$

Therefore, the capacitor voltage at $(k+1)^{th}$ instant can be predicted according to (11). To control the capacitor voltage tightly, the cost function is formulated to

$$J_V = (V_{c\alpha}^{ref} - V_{c\alpha}^{k+1})^2 + (V_{c\beta}^{ref} - V_{c\beta}^{k+1})^2 \quad (12)$$

where $V_{c\alpha}$ and $V_{c\beta}$ are the real and imaginary components of the capacitor voltage vector, respectively. Based on this cost function, the voltage vector that generates the least value of J_V will be applied during the next sampling period. Because the α and β components are tightly controlled, the V_c can track its reference. Thus stable and sinusoidal voltage can be established. Grid synchronization can be achieved simply by setting the voltage reference in cost function (12) as actual grid voltage.

4.2. Model predictive power control in grid-tied mode

Fig. 8(b) illustrates the proposed control method. In grid-tied mode, the control objective of the ac/dc interlinking converter is to maintain a stable dc-bus voltage for dc loads and to exchange power between the microgrid and the utility system. The active power reference is obtained from the outer voltage loop where the PI regulator is used to regulate the dc-bus voltage. When resource conditions or load demand change, the dc-bus voltage is adjusted to constant so as to keep the power balance within the system. Next, the reactive power command is calculated according to the utility grid status as

$$Q_{ref} = \frac{V_g - V_{gN}}{V_{gN}} \times 100\% \times m_{var} \quad (13)$$

where V_g is the actual grid voltage, V_{gN} is the nominal grid voltage, m_{var} is defined as the coefficient of the reactive power compensation. In this way, the microgrid can provide reactive power compensation according to the grid voltage variations. It is necessary to emphasize that the reactive power compensation should be made based on the voltage violation types, i.e., voltage dip or voltage rise. For voltage dip, Q_{ref} is negative for injecting reactive power from microgrids to utility. On the contrary, Q_{ref} becomes positive for absorbing reactive power from utility. In the future smart grid framework with high microgrid penetration, the accurate amount of reactive power supplied by each microgrid should be determined by considering several factors such as grid voltage variations, transmission line R/X ratio, the reactive capability of itself, etc. It is noted that Eq. (13) designed here is just a simplified form to test the potential reactive capability of the proposed hybrid ac/dc microgrid.

After the power references are obtained, a MPC algorithm is developed to control the active and reactive powers. In MPC, the system model is used to predict the behavior of variables at next sampling instant, and a cost function is then employed as the criterion to select the optimum voltage vector of the converter. Finally, the voltage vector that can minimize the cost function is applied in the next sampling period. According to (3)–(5), the active and reactive powers at the end of a sampling period can be predicted as

$$P^{k+1} = T_s \left[-\frac{R}{L} P^k - \omega Q^k + \frac{3}{2L} (|V_g'|^2 - \text{Re}(V_g' V_i^*)) \right] + P^k \quad (14)$$

$$Q^{k+1} = T_s \left[\omega P^k - \frac{R}{L} Q^k - \frac{3}{2L} \text{Im}(V_g' V_i^*) \right] + Q^k \quad (15)$$

where ω is the grid frequency in radians. Since the control objectives in grid-tied mode are the active and reactive powers. Therefore, we can evaluate the effects of each voltage vector on P and Q and then select the one minimizing the following cost function.

$$J_p = (P_{ref} - P^{k+1})^2 + (Q_{ref} - Q^{k+1})^2 \quad (16)$$

In this paper, the proposed predictive controller belongs to the converter-level control. The proposed model predictive voltage and power control scheme is used to select the optimal switching states of the ac/dc converter for regulating active and reactive powers. The prediction step is very short at micro-seconds. Definitely, multiple steps ahead prediction can be employed to obtain better performance such as reduced power ripples [33]. But, longer prediction step will not affect the SOC-oriented charging/discharging scheme. The proposed predictive controller here is just a tool to control the ac/dc interlinking converter. The coordinated control between the ac/dc converter and the buck-boost converter, along with the SOC-oriented charging/discharging scheme, is the key to ensure stable operation and power balance under different cases. Therefore, for the sake of simplicity, single step ahead prediction is employed in the predictive controller.

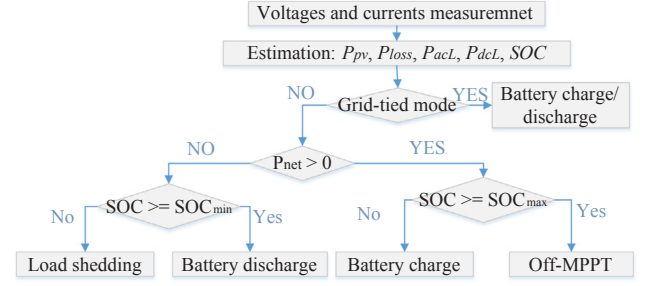


Fig. 9. Energy management system.

5. Energy management system

Two level coordination controls are used to maintain system stable operation. At the system level, the power flow and the operation modes of each converter are determined by the energy management system (EMS) in the control center. The power in the microgrid under various generation and load conditions should be balanced as follows [13]

$$P_{pv} - P_{loss} = P_{acL} + P_{dcL} + P_{bat} \quad (17)$$

where P_{bat} is the battery power and P_{loss} is the total power loss in the microgrid. P_{net} is defined as the total power generation subtracted by load power (P_{acL} and P_{dcL}) and P_{loss} . Fig. 9 shows the logic diagram of the EMS. If the power generated by the PV is insufficient for the power demand at the load side, the battery discharges to provide addition power if its SOC is greater than the minimum value, otherwise load shedding is needed. For example in a residential microgrid network, critical loads include equipment in hospitals and data centers. Non-critical loads could be the electric appliances at home like washing machines and TVs. Once the SOC drops to SOC_min, non-critical loads will be first switched off. If the SOC still hits the bottom at SOC_min, critical loads will then be cut off subsequently. It is worth mentioning that, to achieve economic load shedding, sophisticated load management algorithm is needed. This requires advanced information communication technology with smart meters for real-time load monitoring and dispatching, which is out of the scope of this paper. In case of power generation exceeding the load consumption, the battery is charged to store the surplus power. If the battery SOC exceeds its maximum, Off-MPPT operation is required. Once the powers are dispatched based on the EMS, the individual power converters are controlled coordinately at the local level using the methods proposed in Section III.

6. Case studies

Simulations has been carried out using MATLAB/Simulink based on the microgrid shown in Fig. 1. The microgrid is connected to the 25 kV line through a 260 V/25 kV 2.5MVA transformer, and then further connected to the 120 kV line through a 19-km feeder and a 25 kV/120 kV 47MVA transformer. Table 1 lists the system parameters. The switching frequencies of PV converter and battery converter are 5 kHz. The sampling frequency of the ac/dc interlinking converter is 20 kHz, resulting in about 3.3 kHz average switching frequency.

Table 1
System parameters.

DC grid voltage	1200 V
AC grid voltage	25 kV/0.69 kV
Solar PV	SunPower Spr-305E-WHT-D
Energy storage system	Lithium-ion battery, 0.5 MW, 300 V, 1.3 kA h
DC-link capacitor	6 mF
LC filter	$L_f = 0.6$ mH, $C_f = 1338$ μ F, $R_f = 0.019$ Ω

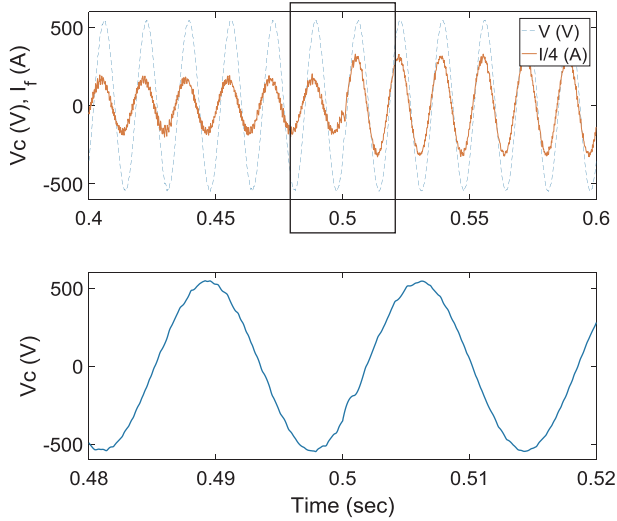


Fig. 10. AC voltage (capacitor voltage of LC filter) and interlinking converter output current with constant PV and variable load at autonomous mode.

6.1. Autonomous operation

In the first test, the solar irradiation level is fixed at 400 W/m^2 that is corresponding to 0.78 MW output. The ac load steps up from 0.5 MW to 1 MW at 0.5 s . Fig. 10 shows the voltage and current responses at the ac subgrid. It can be seen that the interlinking converter automatically increases the output current to meet the load demand, while the ac voltage is maintained stable. Fig. 11 shows the dynamic response of the dc subgrid. To smooth the gap between generation and consumption, the battery is changed from charging to discharging mode after 0.5 s to supply energy. Consequently, the direction of the battery current is reversed and the SOC decreases after 0.5 s . The dc-bus voltage is maintained by the bidirectional buck-boost converter using outer-voltage and inner-current feedback loop.

In the second test, the solar irradiation level ramps up from 400 W/m^2 (0.78 MW) to 700 W/m^2 (1.4 MW). The 0.5 MW ac load is kept connected. A 0.3 MW dc load is switched in at 0.5 s . Fig. 12 shows the performance of the dc subgrid. It can be seen that the bidirectional dc-dc converter is able to maintain the dc-bus voltage and regulate the charging current properly to keep the power balance. In this case, the battery is kept being charged in varying rates as the generation is always greater than consumption.

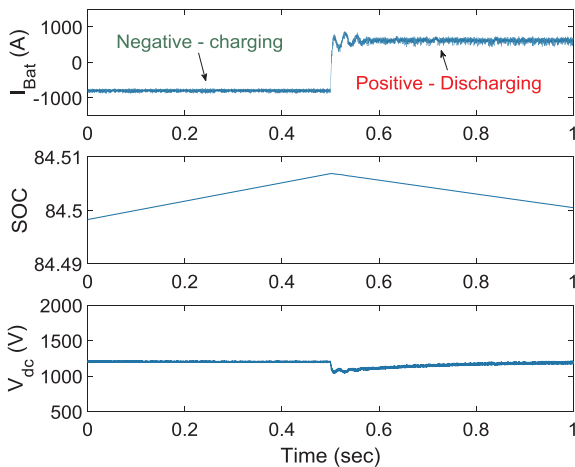


Fig. 11. DC-bus voltage and battery response with constant PV and variable load at autonomous mode.

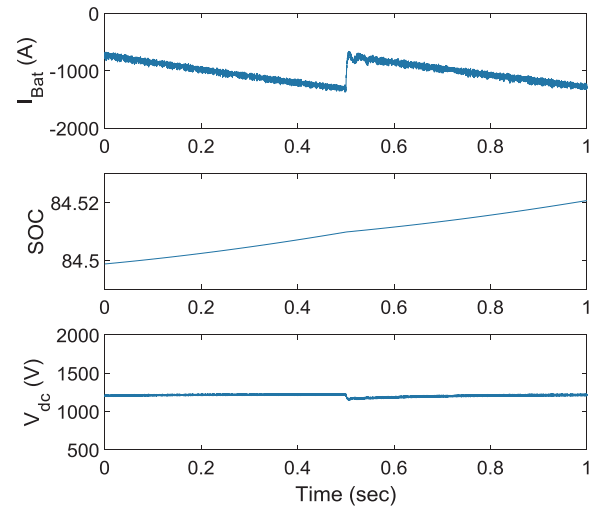


Fig. 12. DC-bus voltage and battery response with variable PV and variable load at autonomous mode.

6.2. Grid synchronization and connection

At autonomous operation, the ac/dc interlinking converter is controlled using model predictive voltage control scheme to establish stable ac voltage. When grid synchronization starts, the voltage reference in cost function (12) is set as the actual grid voltage. Once the ac voltage can match the grid voltage, the microgrid can be connected to the utility grid. The grid synchronization process and connection are presented in Fig. 13. From top to bottom, the waveforms are phase A voltage at ac subgrid terminal, phase A voltage of utility grid, and interlinking converter output currents, respectively. It can be observed that the microgrid voltage can track the grid voltage tightly within 5 ms , prompting smooth grid connection without major overshoot current. After successful grid connection, the power can be transferred between the microgrid and utility.

6.3. Grid-tied operation

At grid-tied mode, the dc-bus voltage is maintained by the interlinking converter based on model predictive power control scheme. The battery charging/discharging current is regulated by the bidirectional dc-dc converter according to the SOC. The solar irradiation is fixed at

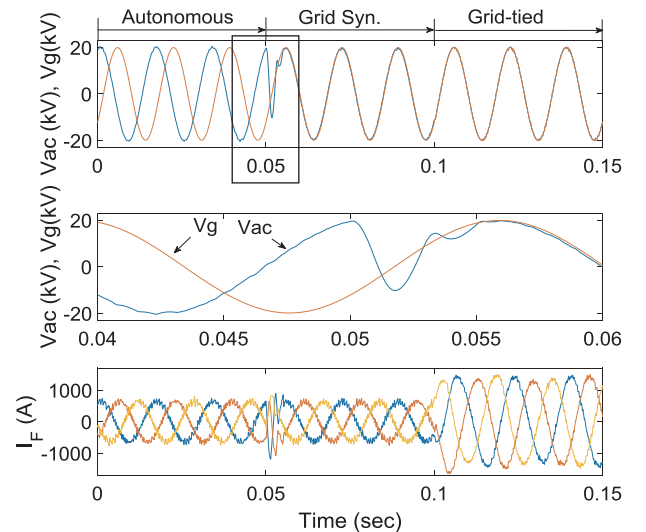


Fig. 13. System response of grid synchronization and grid connection.

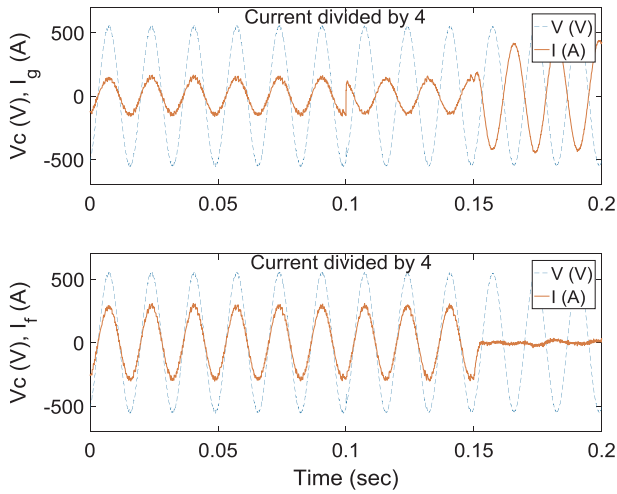


Fig. 14. AC subgrid voltage, utility grid current and ac/dc interlinking converter output current under variable load condition at grid-tied mode.

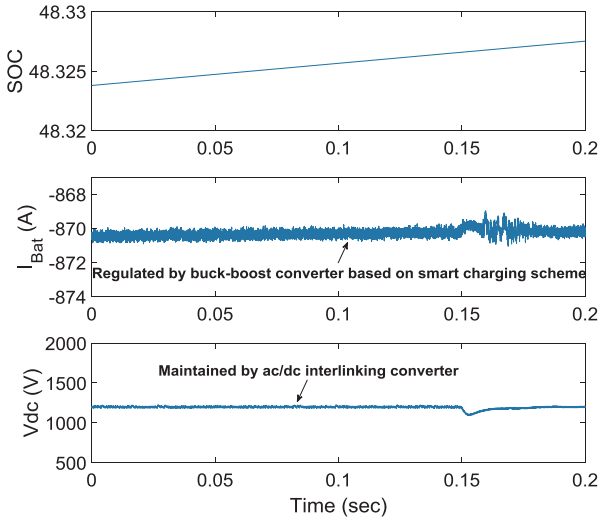


Fig. 15. DC subgrid voltage and battery response under variable load condition at grid-tied mode.

600 W/m² (around 1.2 MW output). A 0.5 MW ac load is kept connected. Another 1 MW ac load is connected at 0.1 s and 1 MW dc load is connected at 0.15 s. Fig. 14 presents the responses at the ac side of the microgrid. It is seen from grid current direction that the power is injected from the utility to the microgrid after 0.1 s because the PV generation is less than the total load demand. It can be also seen that the direction of the interlinking converter output current is reversed after 0.15 s. This means that the power is passed to dc subgrid for the dc load and the battery charging. Fig. 15 shows the responses of the battery and the dc-bus voltage.

6.4. Complete operation under different conditions

For better illustration, a more complex scenario is demonstrated with different operation modes and load conditions. The dc-bus voltage and ac voltage are shown in p.u. in magnitude. Solar irradiation level ramps up from 400 W/m² at 1 s to 800 W/m² at 2 s, then ramps down from 800 W/m² at 3 s to 400 W/m² again at 4 s. Table 2 lists the operation events. Fig. 16 shows the comprehensive overview of the system performance. Observations are summarized and discussed as follows.

Event 1: 0.5 MW ac load is switched in at 0.5 s. The PV output is still

Table 2

Events during operation.

Events	Operations	Time (s)
1	0.5 WM AC load switched in	0.5
2	Solar PV output increases	1
3	0.6 MW DC load switched in	1
4	Grid synchronization	1.4
5	Grid connection	1.6
6	1 MW DC load switched in	2.5
7	Solar PV output decreases	3
8	1 MW DC load switched out	3
9	Grid voltage dips by 10%, $m_{var} = 4 \times 10^6$	3.5

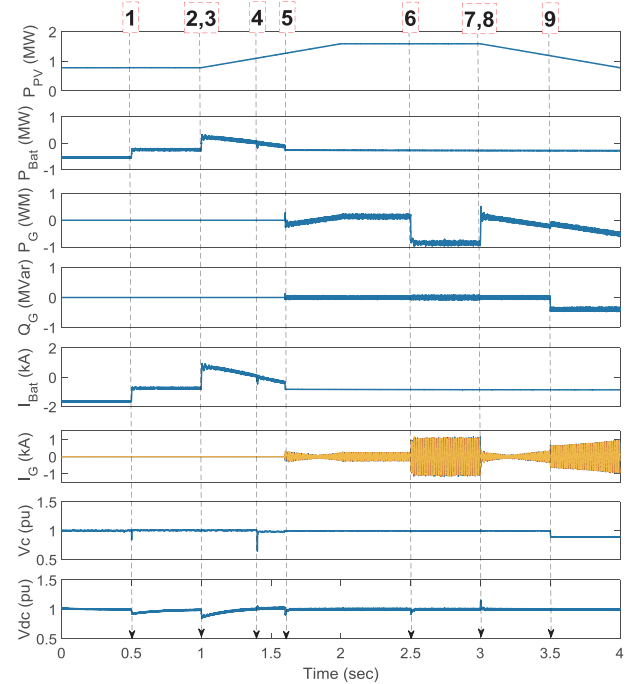


Fig. 16. System performance under variable power generation and load condition at different operation modes.

greater than the load demand, so the battery is kept being charged but in lower charging rate compared with before.

Event 2,3: 0.6 MW dc load is switched in at 1 s and the PV output starts to increase with increased solar irradiation level. At that moment, the total load demand reaches 1.1 MW that is greater than the PV output, leading to the drop of dc-bus voltage. The bidirectional dc-dc converter then responds immediately in the following manner: The outer voltage control loop generates a positive current reference. The inner current control loop adjusts the duty cycle to force the current to flow from the battery to the dc bus, which resulting in discharging of the battery.

Event 4: The grid synchronization begins by setting the voltage reference to the actual grid voltage in the ac/dc interlinking converter controller. The ac subgrid and dc subgrid are stable and operating properly during the synchronization process.

Event 5: The microgrid is connected to the utility grid at 1.6 s. It can be seen that the connection process is smooth without overshoot currents. Since utility power is available, the battery operates in charging mode to store energy afterwards. The charging rate varies with respect to the SOC to achieve a smart charging, as illustrated in Section III. It is worth mentioning that the battery can also operate at discharging mode in grid-tied operation in some circumstance as long as the SOC is greater than SOC_{min} (10%), for example, when the electricity price is relatively expensive at that time of the day.

Event 6: A large load demand is emulated by switching in 1 MW dc load at 2.5 s. It can be seen that, to keep the power balance, extra power is imported from utility grid, as shown by the negative P_G in Fig. 16. During this large load variation, the dc voltage is well controlled by the interlinking converter while the ac voltage is fixed by the utility grid.

Event 7,8: 1 MW dc load is switched out and the PV output starts to reduce as solar irradiation decreases. From 3 s to around 3.2 s, the microgrid exports power to the utility as the power generation is greater than the total powers required by the load and the battery. After 3.2 s, the microgrid starts to import power from the utility because the PV output is not sufficient.

Event 9: 10% utility voltage dip occurs at 3.5 s at the 120 kV transmission line, leading to voltage drop along the feeder and eventually the voltage drop at the microgrid voltage (i.e., the voltage V_c). The reactive power coefficient in (13), m_{var} , is set to 4×10^6 for 0.4 MVar reactive power injection. It is noted that m_{var} should be determined by considering the total capacity of the microgrid and the active power consumption. Due to the direct power control structure and the excellent dynamic response, once the voltage dip is detected, the MPPC controller is able to immediately provide reactive power support. To better observe the voltage profile using reactive power compensation, the zoom-in waveform of V_c amplitude is plotted in Fig. 17. It can be seen that the voltage has been boosted up to around 0.908 pu. It is noted that the reactive power capacity is relatively small compared to the utility grid capacity. Hence, the voltage profile cannot be improved significantly. In the future smart grid with high penetration of microgrids, however, the coordinated reactive power compensation among these microgrids can play an important role to grid voltage support.

6.5. Microgrid operation under practical solar irradiation

To verify the proposed method with practical consideration, the real-world solar irradiation profile on 05-January-2018 is used for generating PV output, which is plotted in Fig. 18. On the demand side, a 0.3 MW ac load is kept connected. A 0.3 MW dc load is switched in at 5 s and a 0.4 MW ac load is switched in at 7 s. In order to observe a faster change of SOC, the battery capacity has been reduced to 13 A h. The microgrid performance in autonomous mode under such fluctuating solar irradiation is presented in Fig. 19. As can be seen, the BES charges or discharges properly to fill the power gap between the generation and consumption, which can be demonstrated by the battery current and

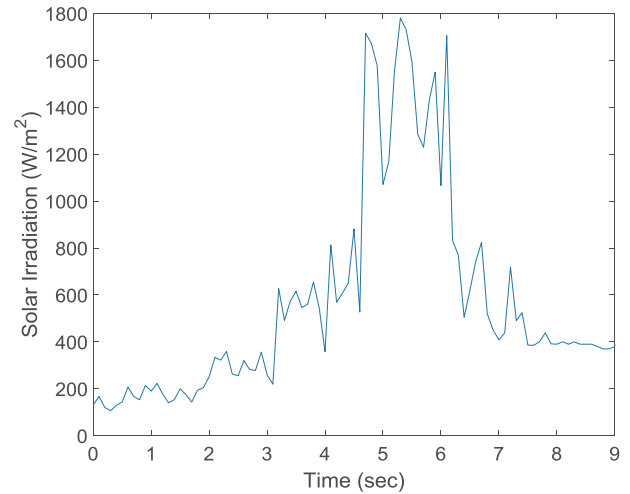


Fig. 18. Real-world solar irradiation.

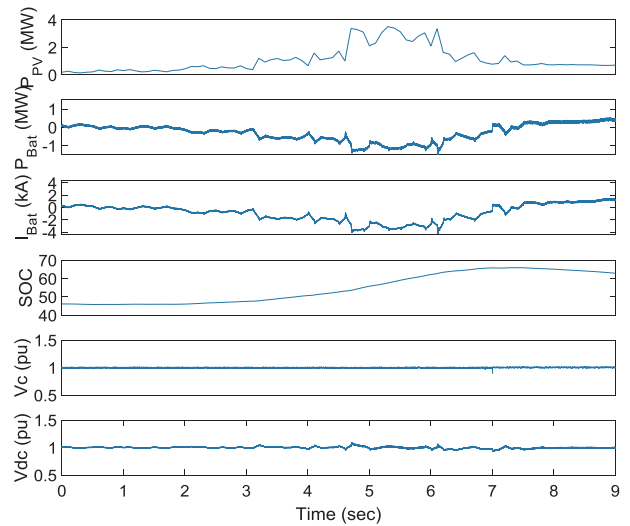


Fig. 19. Microgrid performance under real-world solar irradiation.

the SOC. The dc-bus voltage is maintained by the BES using outer-voltage inner-current double loop control, while stable ac voltage is generated by the ac-dc interlinking converter using MPVC scheme.

7. Conclusion

In this paper, a model predictive power and voltage control (MPPVC) strategy has been proposed for a hybrid ac/dc microgrid. The main contribution of MPPVC is to enable stable operation of microgrids. In addition, the reactive power capacity of the microgrid has been exploited where flexible active and reactive powers sharing is essential between microgrids for high power quality and system stability. Proper charging and discharging of energy storage have also been demonstrated in both grid-tied and autonomous modes under variable power generation and variable load conditions. Comprehensive tests have validated the proposed microgrid architecture and the associated control, showing great potential applications in smart grid development.

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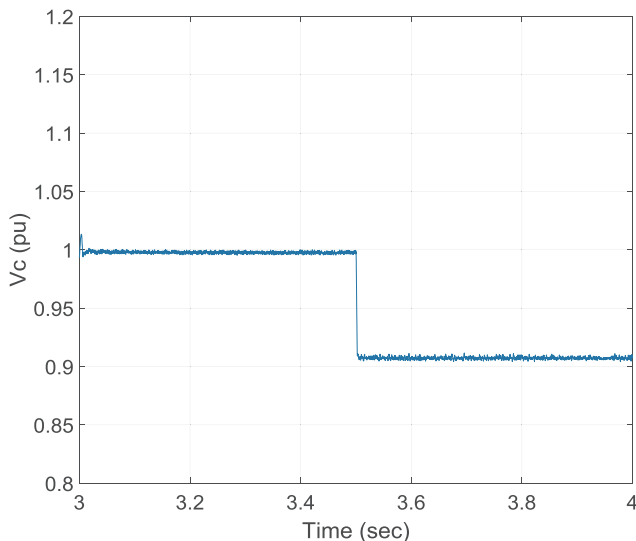


Fig. 17. Microgrid ac voltage profile with reactive power compensation under 10% utility voltage dip at 120 kV transmission line.

Conflict of interest

None.

References

- [1] Wang C, Yan J, Marnay C, et al. Distributed energy and microgrids (DEM). *Appl Energy* 2018;210:685–9.
- [2] Lasseter RH. Smart distribution: coupled microgrids. *IEEE Proc June* 2011;99(6):1074–82.
- [3] Olivares DE, et al. Trends in microgrid control. *IEEE Trans Smart Grid July* 2014;5(4):1905–19.
- [4] Guerrero JM, Vasquez JC, Matas J, Vicuna LG, Castilla M. Hierarchical control of droop-controlled ac and dc microgrids – a general approach toward standardization. *IEEE Trans Ind Electron Jan.* 2011;58(1):158–72.
- [5] Urtasun A, Sanchis P, Marroyo L. State-of-charge-based droop control for stand-alone ac supply systems with distributed energy storage. *Energy Convers Manage* 2015;106:709–20.
- [6] Makrygiorgou DI, Alexandridis AT. Distributed stabilizing modular control for stand-alone microgrids. *Appl Energy* 2018;210:925–35.
- [7] Nejabatkhah F, Li YW. Overview of power management strategies of hybrid ac/dc microgrid. *IEEE Trans Power Electron Dec* 2015;30(2):7072–89.
- [8] Ratnam EL, Weller SR, Kellett CM. Scheduling residential battery storage with solar PV: assessing the benefits of net metering. *Appl Energy* 2015;155:881–91.
- [9] Kaabeche A, Ibtouen R. Techno-economic optimization of hybrid photovoltaic/wind/diesel battery generation in a stand-alone power system. *Appl Energy* 2014;103:171–82.
- [10] Kim M, Bae S. Decentralized control of a scalable photovoltaic (pv)-battery hybrid power system. *Appl Energy* 2017;188:444–55.
- [11] Merabet A, Ahmed KT, Ibrahim H, Beguenane R, Ghias A. Energy management and control system for laboratory scale microgrid based wind-pv-battery. *IEEE Trans Sustain Energy Jan.* 2017;8(1):145–54.
- [12] Ahadi A, Kang S-K, Lee J-H. A novel approach for optimal combinations of wind, pv, and energy storage system in diesel-free isolated communities. *Appl Energy* 2016;170:101–15.
- [13] Liu X, Wang P, Loh PC. A hybrid AC/DC microgrid and its coordination control. *IEEE Trans Smart Grid June* 2011;2(2):278–86.
- [14] Ma T, Cintuglu MH, Mohammed OA. Control of hybrid ac/dc microgrid involving energy storage and pulsed loads. *IEEE Trans Ind Appl* 2017;53(1):567–75.
- [15] Li Z, Xu Y. Optimal coordinated energy dispatch of a multi-energy microgrid in grid-connected and islanded modes. *Appl Energy* 2018;210:974–86.
- [16] Mahmud N, Zahedi A, Mahmud A. A cooperative operation of novel pv inverter control scheme and storage energy management system based on anfis for voltage regulation of grid-tied pv system. *IEEE Trans Ind Inform* 2017;13(5):2657–68.
- [17] Li X, Hui D, Lai X. Battery energy storage station (bess)-based smoothing control of photovoltaic (pv) and wind power generation fluctuations. *IEEE Trans Sustain Energy* 2013;4(2):464–73.
- [18] Rahimi-Eichi H, Baronti F, Chow M-Y. Online adaptive parameter identification and state-of-charge coestimation for lithium-polymer battery cells. *IEEE Trans Ind Electron* 2014;61(4):2053–61.
- [19] Wang L, Wang L, Liao C. Research on improved EKF algorithm applied on estimate EV battery SOC. In: *Proc. of Asia-Pacific Power and Energy Engineering Conf.*; 2010. p. 1–4.
- [20] Wei Z, Zou C, Leng F, Soong BH, Tseng K-J. Online model identification and state-of-charge estimate for lithium-ion battery with a recursive total least squares-based observer. *IEEE Trans Ind Electron* 2018;65(2):1336–46.
- [21] Luo L, et al. Optimal siting and sizing of distributed generation in distribution systems with pv solar farm utilized as STATCOM (PV-STATCOM). *Appl Energy* 2018;210:1092–100.
- [22] Kanchanaharuthai A, Chankong V, Loparo KA. Transient stability and voltage regulation in multimachine power systems Vis-à-vis STATCOM and battery energy storage. *IEEE Trans Power Syst Sep.* 2015;30(5):2404–16.
- [23] Turitsyn K, Sulc P, Backhaus B, Chertkov M. Distributed control of reactive power flow in a radial distribution circuit with high photovoltaic penetration. In: *Proc. of IEEE power and energy society general meeting, USA*; July 2010. p. 1–6.
- [24] Kabir MN, Mishra Y, Ledwich G, Dong ZY, Wong KP. Coordinated control of grid-connected photovoltaic reactive power and battery energy storage systems to improve the voltage profile of a residential distribution feeder. *IEEE Trans Ind Inform* May 2014;10(2):967–77.
- [25] Parisio A, Rikos E, Tzamalidis G, Glielmo L. Use of model predictive control for experimental microgrid optimization. *Appl Energy* 2014;115:37–46.
- [26] Parisio A, Rikos E, Glielmo L. A model predictive control approach to microgrid operation optimization. *IEEE Trans Control Syst Technol Sep.* 2014;22(5):1813–27.
- [27] Wang T, Kamath H, Willard S. Control and optimization of grid-tied photovoltaic storage systems using model predictive control. *IEEE Trans Smart Grid Mar.* 2014;5(2):1010–7.
- [28] Xie S, He H, Peng J. An energy management strategy based on stochastic model predictive control for plug-in hybrid electric buses. *Appl Energy* 2017;196:279–88.
- [29] Ouammi A, Dagdougui H, Dessaint L, Sacile R. Coordinated model predictive-based power flows control in a cooperative network of smart microgrids. *IEEE Trans Smart Grid* 2015;6(5):2233–44.
- [30] Ropp ME, Gonzalez S. Development of a MATLAB/Simulink model of a single-phase grid-connected photovoltaic system. *IEEE Trans Energy Conv* 2009;24(1):195–202.
- [31] Sera D, Mathe L, Kerekes T, Spataru SV, Teodorescu R. On the perturb-and-observe and incremental conductance MPPT methods for PV systems. *IEEE J Photovolt* 2013;3(3):1070–8.
- [32] Tremblay O, Dessaint LA, Dekkiche AI. A generic battery model for the dynamic simulation of hybrid electric vehicles. In: *Proc. of IEEE Veh. Power Propulsion Conf.* p. 284–89.
- [33] Hu J, Zhu J, Dorrell DG. Model predictive control of grid-connected inverters for PV systems with flexible power regulation and switching frequency reduction. *IEEE Trans Ind Appl* 2015;51(1):587–94.